

Project 2: Numerical Solutions to Incompressible Navier–Stokes Equations

Jaden Mecham^{a)}

Department of Mechanical Engineering, University of Nevada, Reno

(Dated: 14 April 2026)

abstract here.

I. INTRODUCTION

The objective of this project is to develop a numerical solver for the incompressible Navier–Stokes equations, which govern the flow of viscous fluids. We will implement a fractional step method with 2nd order Adams–Bashford time integration for the advection term and Crank–Nicolson for the viscous term. We also compare the results to the results reported in the paper by Ghia et al. For this project, we will focus on the lid-driven cavity flow problem, which is a classic benchmark case for validating numerical solvers for incompressible flows.

II. GOVERNING EQUATIONS AND BOUNDARY CONDITIONS

The governing equations for incompressible flow are the Navier–Stokes equations, which consist of the continuity equation and the momentum equations. The continuity equation ensures mass conservation and is given by:

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

where $\mathbf{u}$ is the velocity vector. The momentum equations describe the conservation of momentum and are given by:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u} \quad (2)$$

where p is the pressure, ν is the kinematic viscosity, and t is time. For the lid-driven cavity problem, we consider a square domain with the following boundary conditions:

- No-slip boundary condition on the bottom and side walls: $\mathbf{u} = 0$
- Moving lid at the top wall: $\mathbf{u} = (U, 0)$, where U is the lid velocity
- Initial condition: $\mathbf{u} = 0$ everywhere at $t = 0$
- Reynolds number defined as $Re = \frac{UL}{\nu}$, where L is the length of the cavity

III. OPERATORS

The following operators are used in the numerical method to discretize the governing equations:

III.A. Gradient Operator ($\mathbf{G}$)

The gradient operator is discretized using central differences for the interior points and one-sided differences

for the boundary points. It maps pressure difference to velocity face values using the following formulas:

$$(\mathbf{G}p)_{i,j}^u = \frac{p_{i,j} - p_{i-1,j}}{\Delta x}, \quad (\mathbf{G}p)_{i,j}^v = \frac{p_{i,j} - p_{i,j-1}}{\Delta y} \quad (3)$$

III.B. Divergence Operators ($\mathbf{D}$ and $\mathbf{bc}_D$)

The divergence operator is discretized using central differences for the interior points and one-sided differences for the boundary points. It maps velocity face values to cell-centered divergence. The central points are calculated using the following formula:

$$(\mathbf{D}\mathbf{u})_{i,j} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} + \frac{v_{i,j+1} - v_{i,j}}{\Delta y} \quad (4)$$

The full divergence operator is split into the sum of the interior and boundary contributions. The boundary contribution at each wall is given in the following table:

III.C. Laplacian Operators ($\mathbf{L}$ and $\mathbf{bc}_L$)

For the interior points:

$$(\mathbf{L}\mathbf{u})_{i,j} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2} \quad (5)$$

III.D. Advection Operator ($\mathbf{A}$)

III.E. Conjugate Gradient Solver

III.F. Unit Testing

IV. NUMERICAL METHOD

IV.A. Grid and Discretization

We use a uniform Cartesian grid with $N \times N$ cells to discretize the domain and apply a staggered grid arrangement for the velocity and pressure fields.

V. DISCUSSION

VI. CONCLUSION

^{a)}Electronic mail: jadenmecham@unr.edu